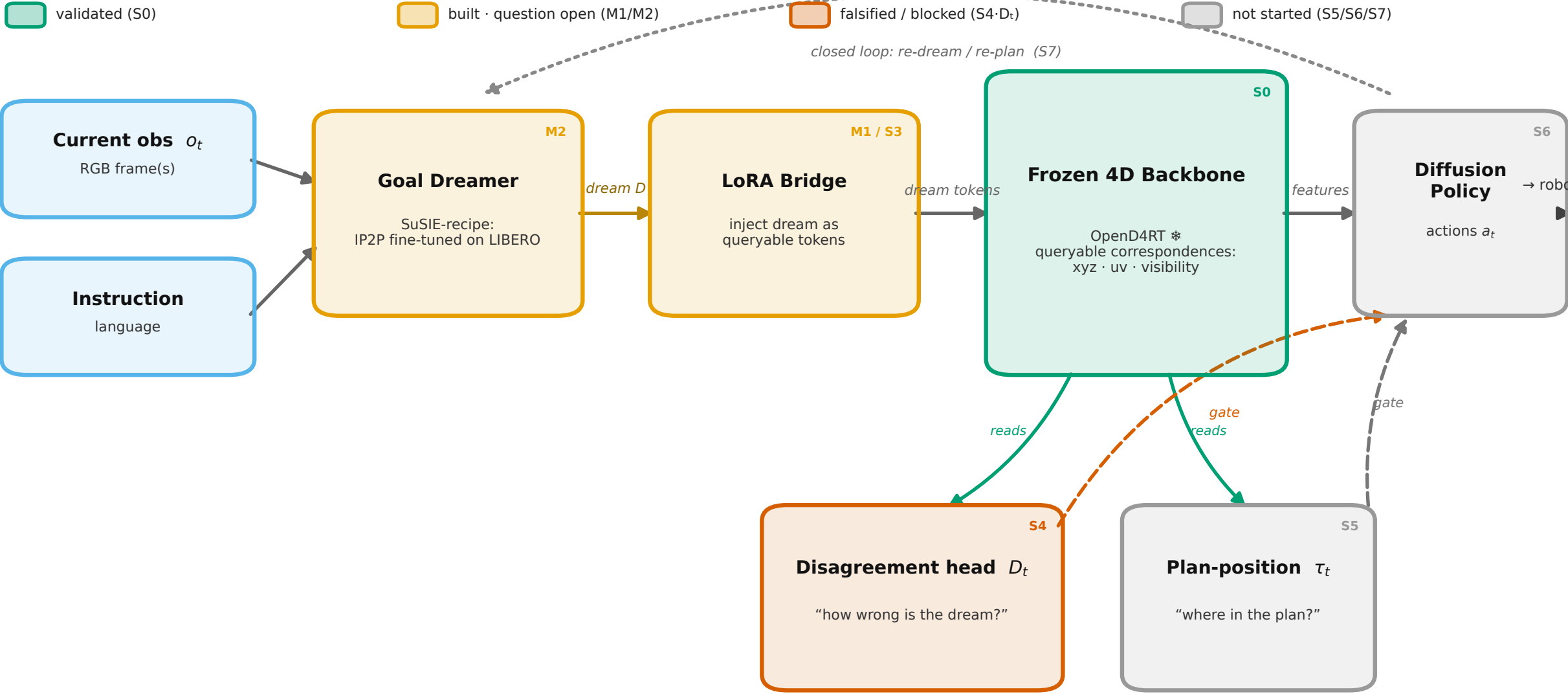


GD-4D — Grounded Dreaming on a Frozen 4D Backbone

goal-image-conditioned robot policy; the dreamed goal is injected into a queryable 4D reconstruction, and two gates (D_t , τ_t) modulate the policy



✗ D_t is **FALSIFIED** as a geometric read-out of the frozen backbone — the current crossroads

Three converging de-risks: M1 premise (a dream’s own self-consistency is identical for real / hallucinated / copied goals) · M2 (on a real dreamer, confound-free geometric delta AUROC 0.51 = chance) · M3 (reference-free cross-consistency ≈ 1.0 on geometrically-broken goals but ~ 0.60 on a plausible-but-wrong future). The backbone finds coherent correspondences for ANY realistic image → geometry cannot answer “is this the CORRECT dream”. $\Rightarrow D_t$ must be LEARNED (a critic) or sourced elsewhere (dreamer / policy uncertainty).